EXAMPLE 14

Friction Pendulum Base-Isolated 3D Frame - Nonlinear Time History Analysis

Problem Description

This is a two-story, three-dimensional frame with base isolation using friction pendulum base isolators. The structure is subjected to earthquake motion in two perpendicular directions using the Loma Prieta acceleration records.

Friction pendulum type base isolators of the type described in Zayas and Low (1990) are modeled using the ETABS ISOLATOR2 elements.

It is important for these isolator elements that the axial load from other loads be modeled before starting the nonlinear analysis. This is achieved by using a factor of unity on the dead load (self weight) on the structure in the nonlinear analysis initial conditions data.

Geometry, Properties and Loads

The structure is modeled as a single reinforced concrete frame with nine column lines and twelve bays. The floor slab is taken to be 8 inches thick, covering all of the specified floor bays at the base and the 1st story level. At the second story level, the corner column and the two edge beams are eliminated, together with the floor slab, to render this particular level anti-symmetric, as depicted in Figure 14-1.

The isolator properties are defined as follows:

Stiffness in direction 1	1E3
Stiffness in directions 2 and 3	1E2
Coefficient of friction at fast speed	.04
Coefficient of friction at slow speed	.03
Parameter determining the variation of the coefficient of friction with velocity	20
Radius of contact surface in directions 2 and 3	60

A modulus of elasticity of 3000 ksi is used. The self-weight of concrete is taken as 150 pcf. Kip-inch-second units are used.

The geometry of the structure is shown in Figure 14-1.

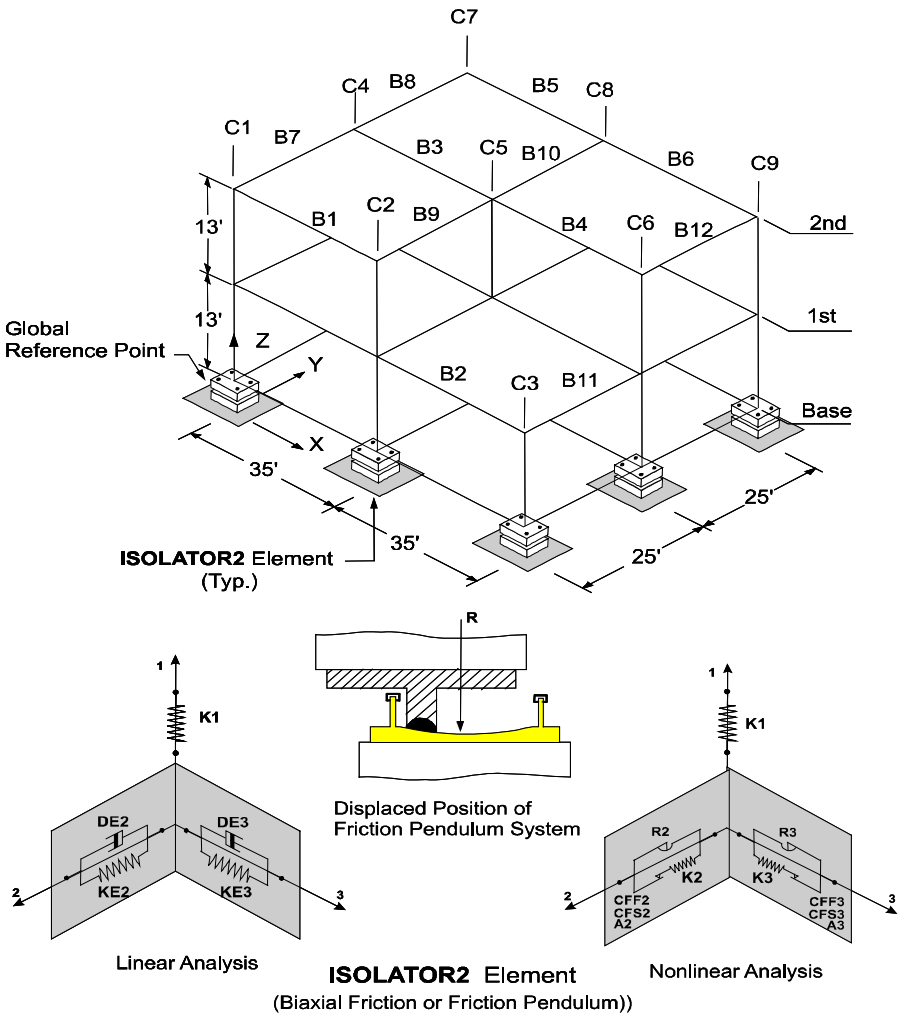


Figure 14-1 Base-Isolated Three-Dimensional Frame

Technical Features of ETABS Tested

- Three-dimensional frame analysis
- Use of area (floor) objects
- Use of biaxial friction pendulum elements
- Point assignments
- Nonlinear time history analysis using ritz vectors

Results Comparison

The example frame analyzed using ETABS is also analyzed using SAP2000 (Computers and Structures 2002) for time history loads (SAP2000 has been verified independently). A comparison of key ETABS results with SAP2000 is presented in Table 14-1.

Table 14-1 Comparison of Result for Time History Analysis

Quantity	ETABS	SAP2000
Maximum Uy Displacement, Column C9 at 2nd Floor	4.2474	4.2069
Maximum Axial Force, Column C1 at base	37.13	38.25

A typical output produced by the program is shown in Figure 14-2. It shows the variation of the displacement of the second story at column line 1.

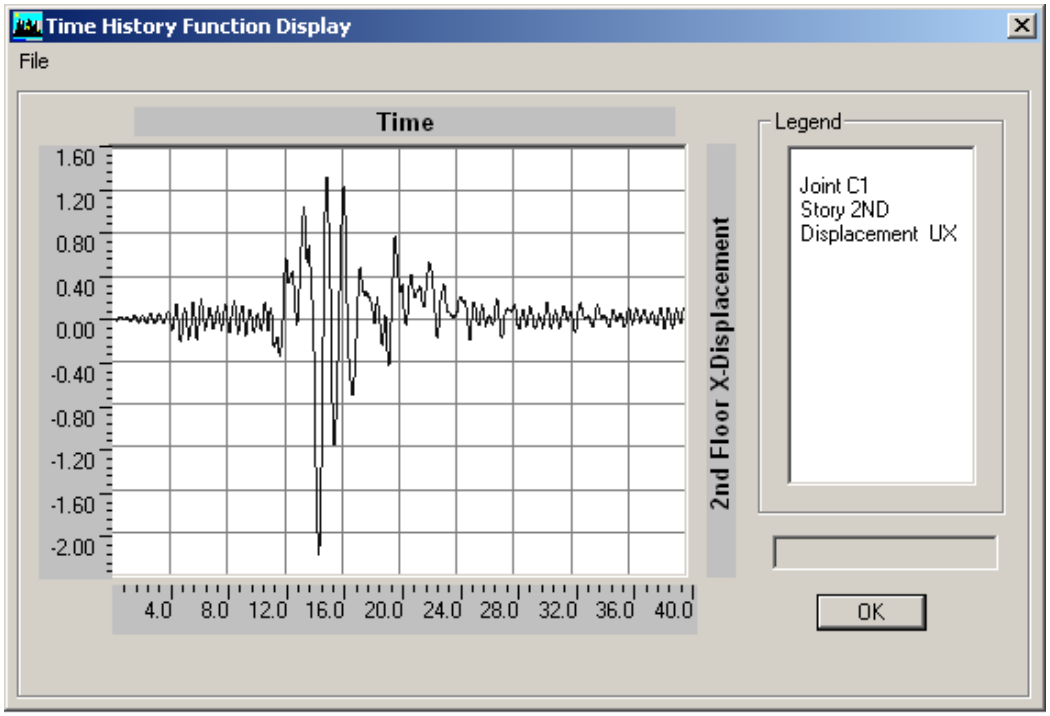


Figure 14-2 Variation of Displacement

Computer Files

The input data file for this example is Example 14.EDB. The time history files are LP-TH0 and LP-TH90. All of the files are provided as part of the ETABS installation.

Conclusion

The results comparison shows acceptable agreement between ETABS and SAP2000.